Hidden States in Macro-Finance:
Why Latent Variables Matter and How to Estimate Them

Introduction

A **latent variable** is an **unobserved quantity** you assume exists because it **helps explain patterns** in what you do observe

Example: **credit cycle, business cycle, financial stress** etc. – this **can not be measured** but is **inferred** from observed data

Formalism : $y_t \approx \Lambda f_t + \varepsilon_t$ *aa*

y_t *: observed vector(s)*

f_t *: latent factors(s)/state(s)*

Λ *: loadings mapping latent -> observed*

ε_t *: idiosyncratic noise*

Loading: is a coefficient telling you how strongly each observed variable "responds to" or "is explained by" a factor

e.g. loading is each variable's sensitivity (beta) to that latent variable ('s)

There cloud be more that one latent factor in a set of variables

Latent variables help in creating structure:

- Dimensionality reduction(sometimes)
- Noise reduction / signal extraction
- Improve interpretability – if the latent factor(s) are constructed with economic meaning
- Improve prediction – they summarise co-movement, evolve smoothly with explicit dynamics

Some Latent Methods

General

- Pure dimensionality reduction : PCA(cross sectional), Factor analysis, autoencoders
- State estimation : Kalman filter states (optimize time-evolving hidden signal + uncertainty)
- Cycle extraction : SSA, MSSA components (optimize decomposition into oscillatory modes)
- Dynamic Factor Models
- Supervised latent representation - Latent variables optimized to predict a target

All produce “latent” objects, but with different meanings and guarantees

Time Series Latent Sates - x2 version possible

1. Filtered state : $f_t \mid y_{1:t}$ *Point in time*
2. Smoothed state: $f_t \mid y_{1:T}$ *Hindsight*

Laten variables can have specific use in analytics e.g.:

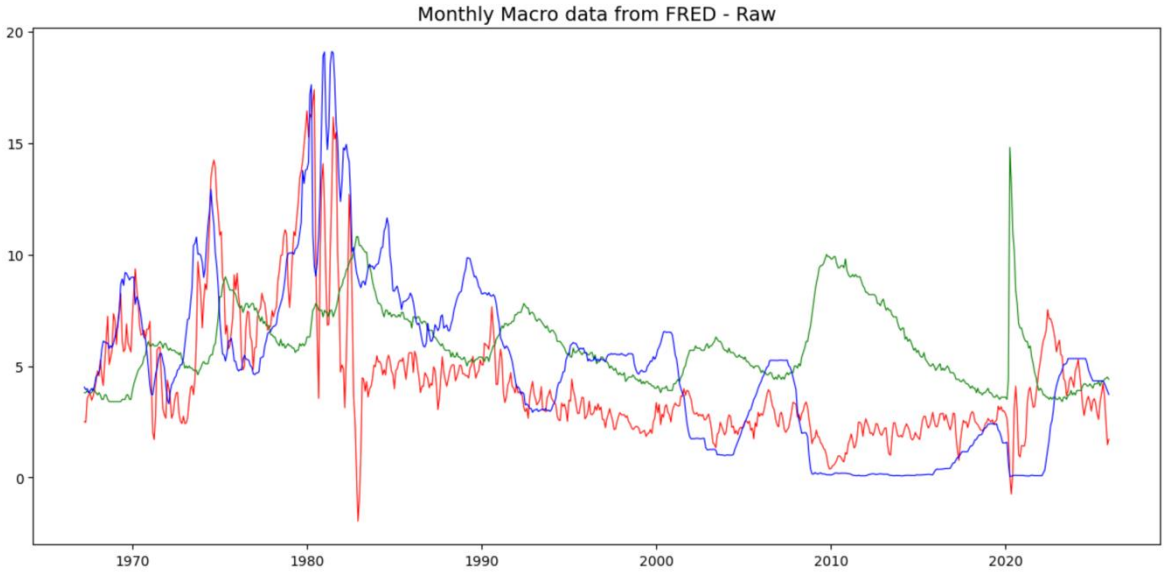
- Replace many correlated variables with few latent factors - compression
- Estimate latent state e.g. business cycle, credit cycle etc. - stabilize
- Improve interpretability – factors map to meaningful concepts
- Other…

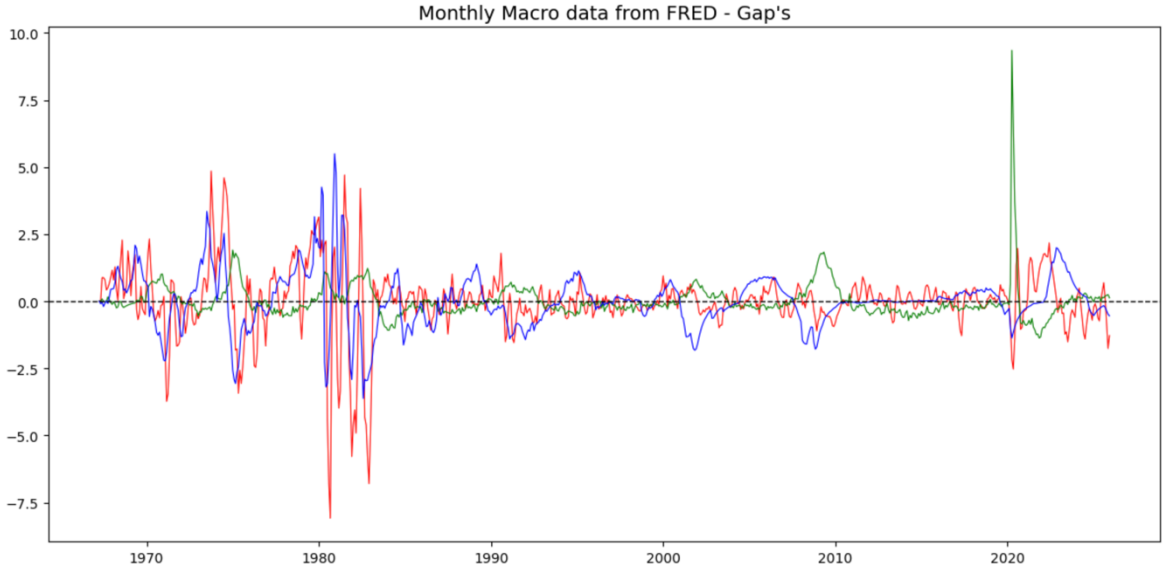
Example 1 – MSSA – Estimate Business and Credit Cycle

Input – Daily Time Series:

- Liquidity
- Unemployment
- Interest rate

- Smoothed using one sided smoother : EWM(12)





Example 1 – MSSA – Estimate Business and Credit Cycle

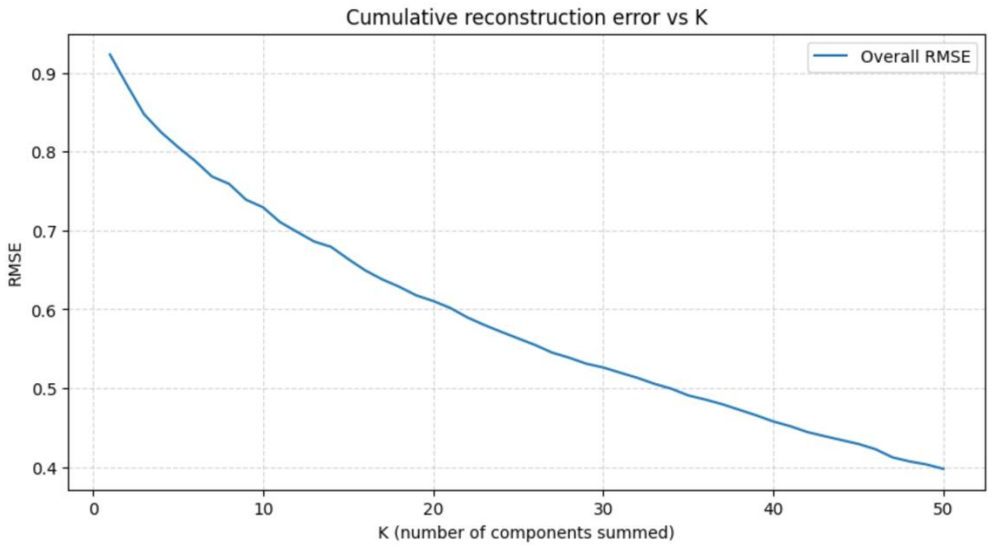
Run MSSA:

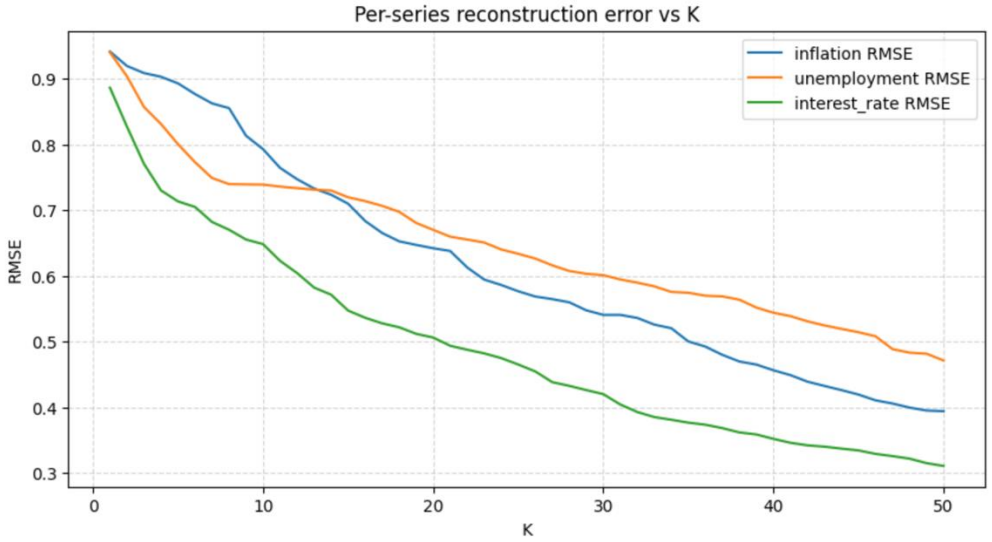
- n-Components = 50
- Window_size = 120 (10 years)

- Output:
  - Reconstructed Components array
    - X shape: (704, 3)
    - components shape: (**3**, 704, 50)
    - 3 series
    - 704 observations
    - 50 frequency components
  - Rank - array([[0, 1, 0],
    [1, 0, 1],
    [9, 2, 2],
    [2, 3, 3],
    [8, 4, 6] etc.

- **No sharp signal/noise separation**: the series contain structure across many time scales; "noise" isn't isolated

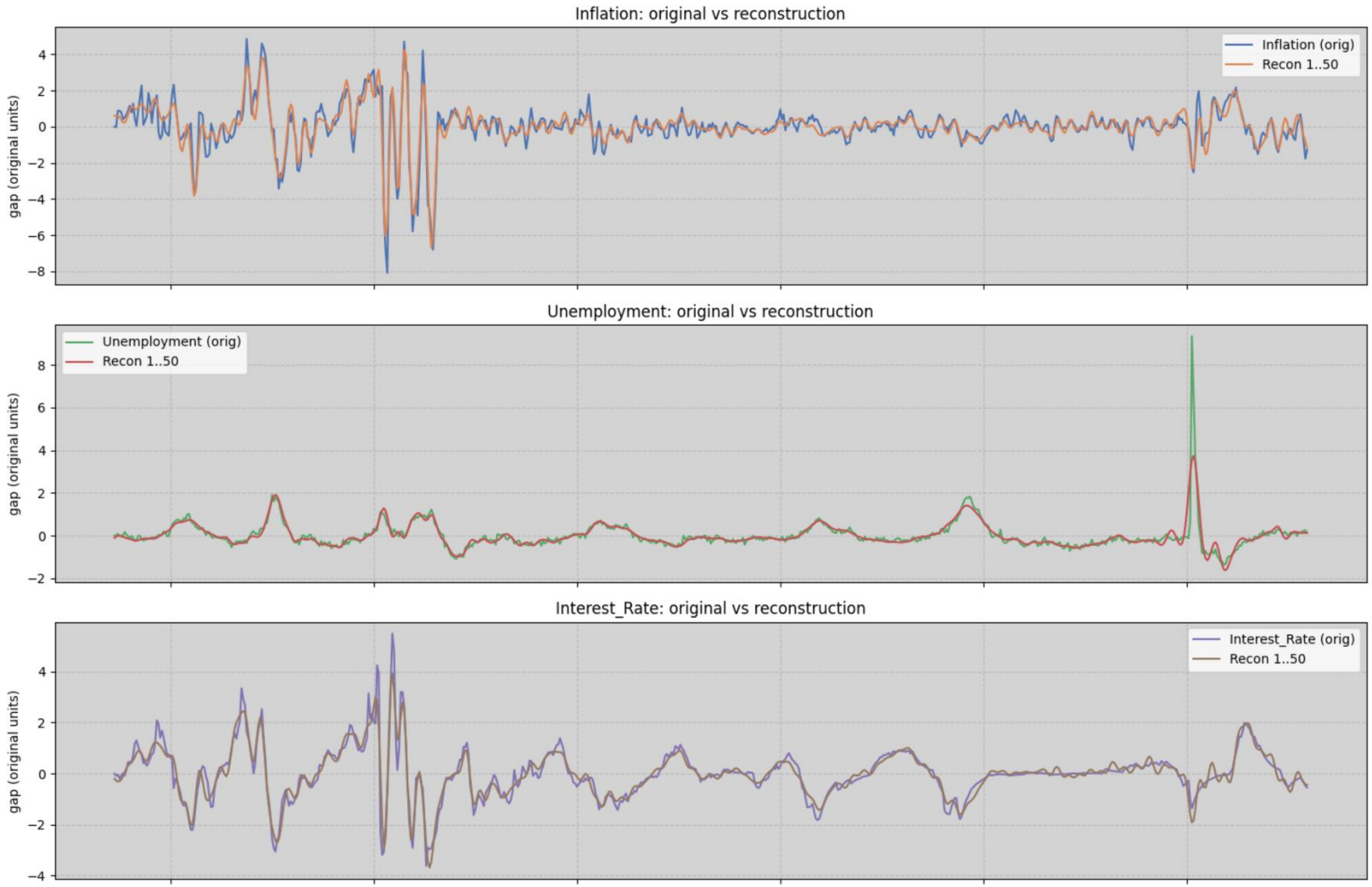
Cumulative Reconstruction





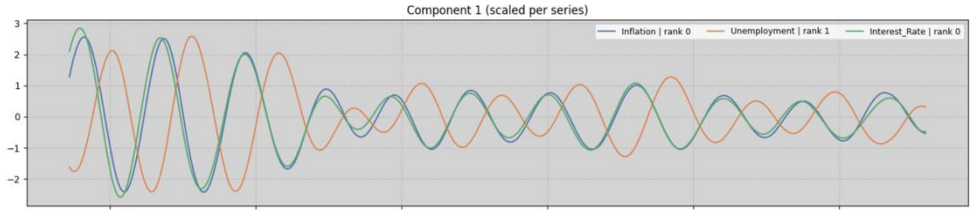
Example 1 – MSSA – Estimate Business and Credit Cycle

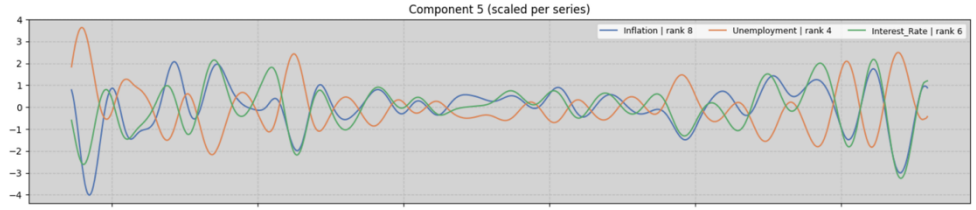
Reconstruction

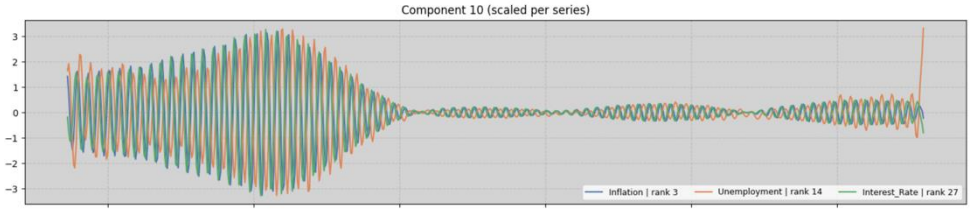


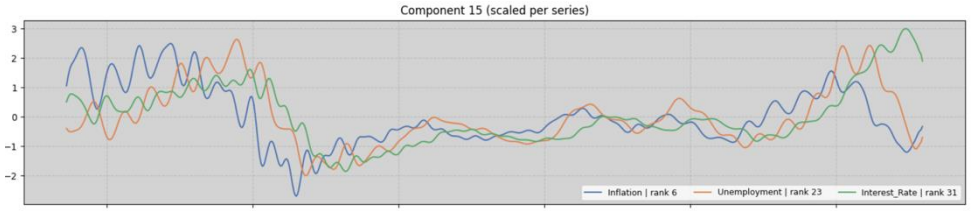
Example 1 – MSSA – Estimate Business and Credit Cycle

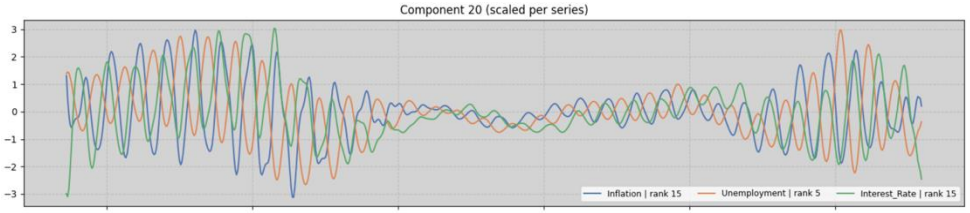
Components x50 - sample

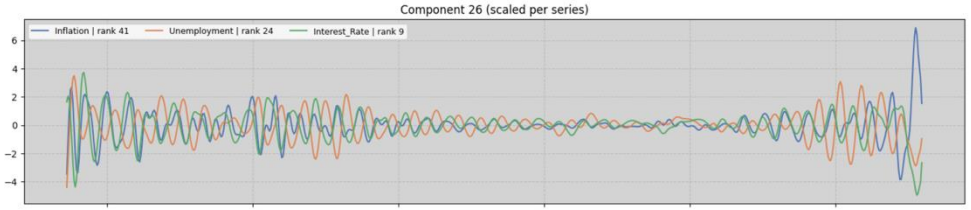


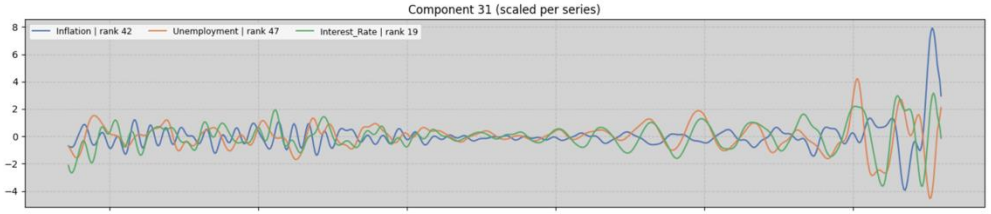


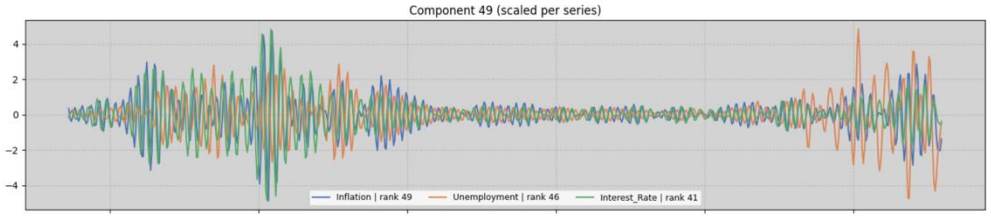






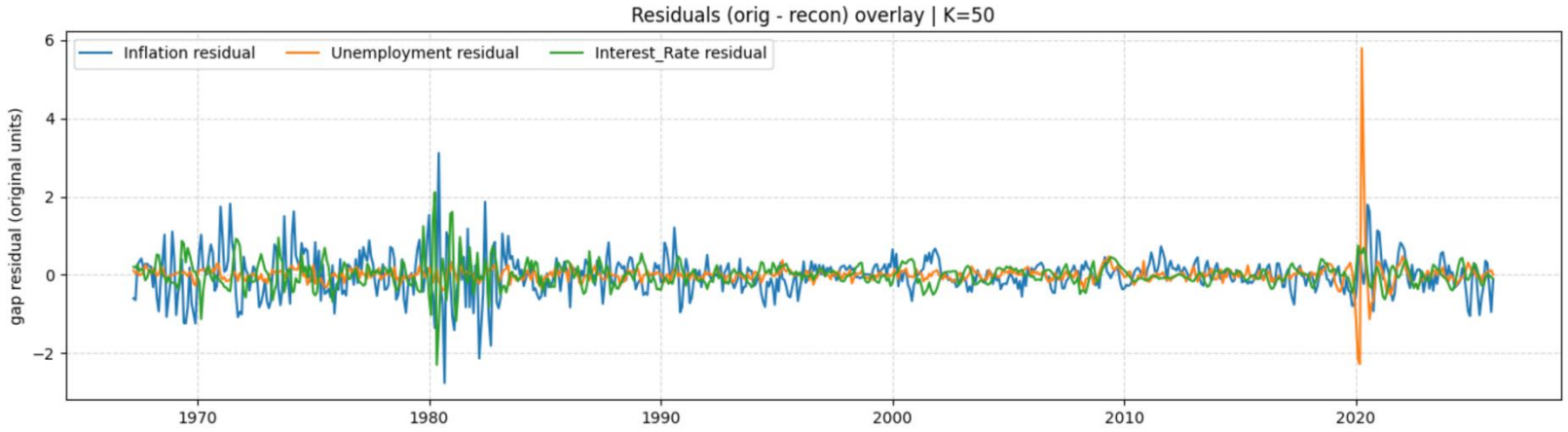






Example 1 – MSSA – Estimate Business and Credit Cycle

Residuals



Example 1 – MSSA – Estimate Business and Credit Cycle

- Business and credit cycles are **persistent, multi-year oscillations** in macro/financial variables

- Estimating their **dominant periods** helps you isolate cycle bands e.g.:
  - business, credit cycle bands etc.
  - build **cycle factors** for regime detection or early warning
  - avoid mixing short-term noise with structural cycles

- Dominant period: the period length (months) where a series has the highest oscillatory power.
  - Tech: the peak in the Fourier power spectrum within a chosen band (e.g., 12–240 months)

- In MSSA/SSA, dominant cycles often appear in pairs because a cycle has two degrees of freedom: amplitude + phase

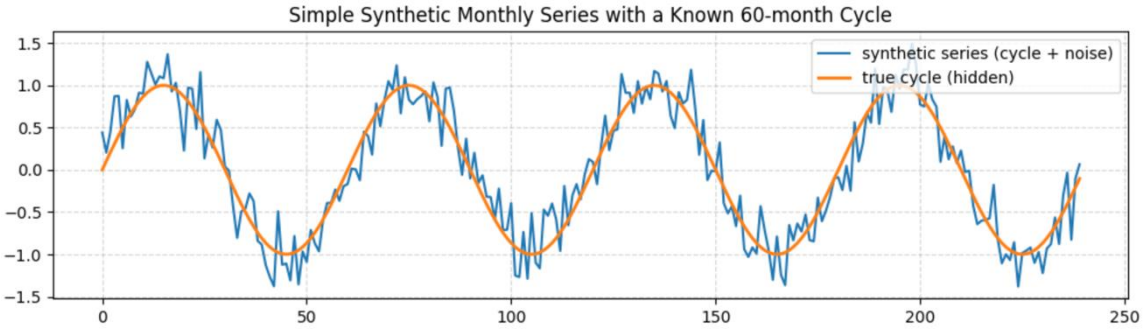
- A single frequency ω can be represented by two basis waves: sine and cosine (same period, phase-shifted)

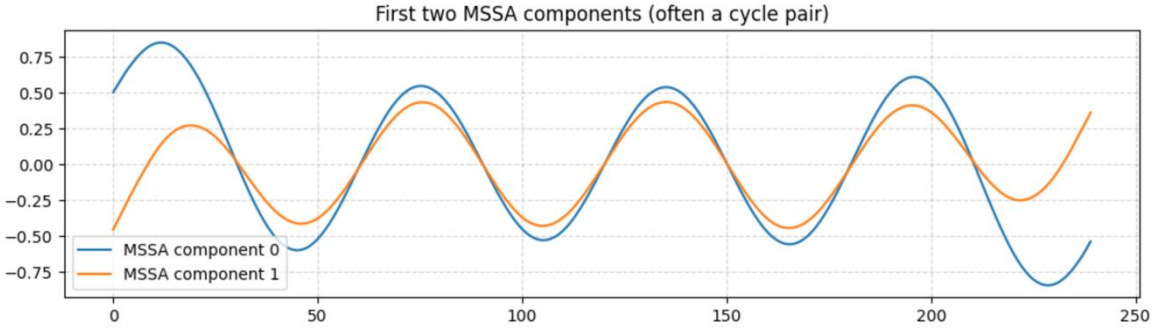
- MSSA/SSA builds a lagged trajectory matrix and applies SVD; an oscillation typically emerges as a rank-2 (two-component) mode

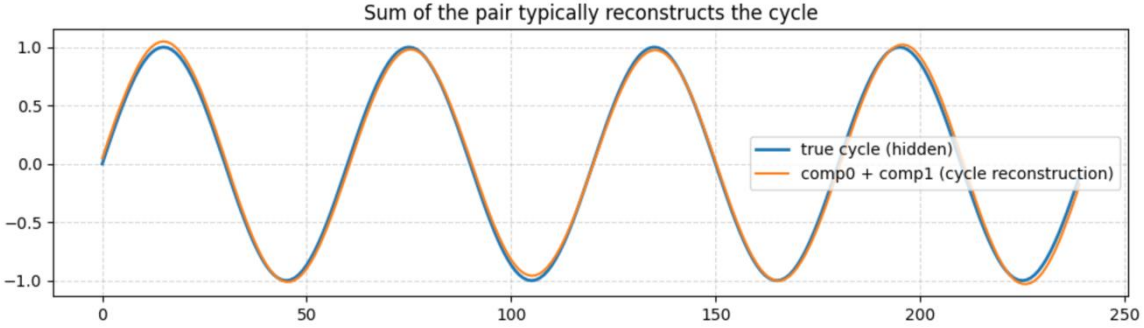
- Therefore, components j and j+1 often share the same dominant period, but represent different phase views of the same underlying cycle.

- **Ex Ante business assumptions Target bands** :
  - business ~24–96 months,
  - credit ~84–180 months (overlaps allowed)

Example 1 – MSSA – Estimate Business and Credit Cycle







Example 1 – MSSA – Estimate Business and Credit Cycle

Dominant Periods

comp_a	comp_b	dom_period_a_months	dom_period_b_months	pair_period_med_months	cycle_band
1	2	64	64	64	business
3	4	117.333333	58.666667	88	business credit
5	6	41.411765	41.411765	41.411765	business
7	8	32	32	32	business
9	10	16	13.283019	14.641509	
11	12	16	16	16	
13	14	13.538462	13.538462	13.538462	
15	16	176	22.709677	99.354839	credit
17	18	22.709677	22.709677	22.709677	
19	20	176	176	176	credit
21	22	88	14.666667	51.333333	business
23	24	14.666667	70.4	42.533333	business
25	26	14.978723	18.526316	16.75252	
27	28	39.111111	20.705882	29.908497	business
29	30	12.137931	12.137931	12.137931	
31	32	44	39.111111	41.555556	business
33	34	58.666667	234.666667	146.666667	credit
35	36	234.666667	54.153846	144.410256	credit
37	38	50.285714	54.153846	52.21978	business
39	40	14.666667	14.666667	14.666667	

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Example 1 – MSSA – Estimate Business and Credit Cycle

Dominant Business Periods:

- Period Median Between 24 & 96 Months
- Relative difference between adjacent components < 0.2

comp_a	comp_b	dom_period_a_months	dom_period_b_months	pair_period_med_months	cycle_band	rel_diff
7	8	32	32	32	business	0
5	6	41.411765	41.411765	41.411765	business	0
31	32	44	39.111111	41.555556	business	0.11765
37	38	50.285714	54.153846	52.21978	business	0.07407
1	2	64	64	64	business	0

Dominant Credit Periods:

- Period Median Between 84 & 180 Months
- Relative difference between adjacent components < 0.2

comp_a	comp_b	dom_period_a_months	dom_period_b_months	pair_period_med_months	cycle_band	rel_diff
19	20	176	176	176	credit	0

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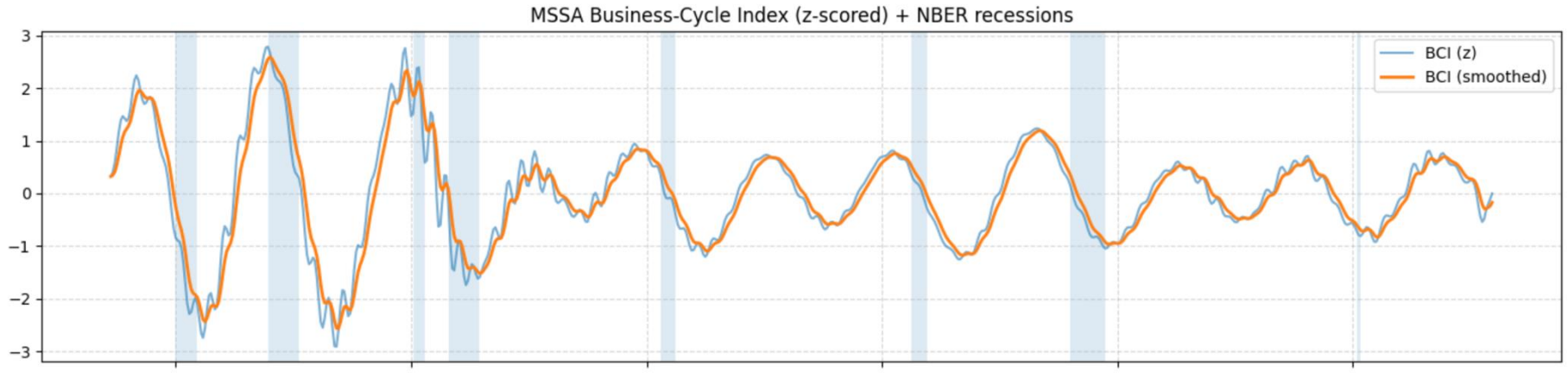
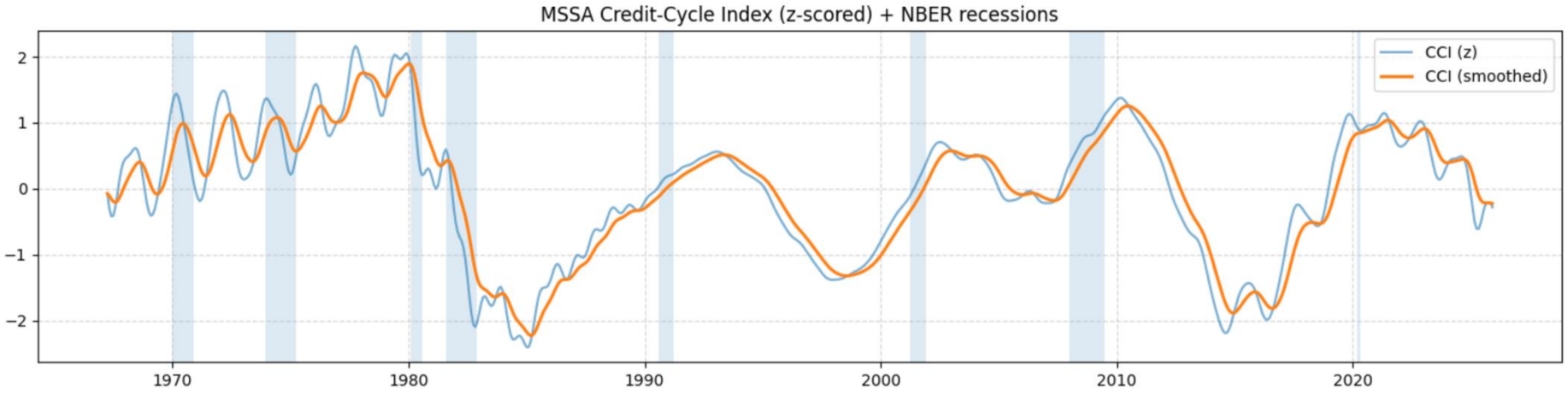
Dominant Business Periods:



Dominant Credit Periods:



NBER - National Bureau of Economic Research

Example 1 – MSSA – Estimate Business and Credit Cycle Summary

- **Business & Credit Cycles** are **Latent variables**
  - Low-dimensional, interpretable state variables extracted from many macro/market series
  - Improve signal quality and reduce multicollinearity / feature sprawl

- **Downstream analytics (Examples):**
  - **Regime & recession analytics** (cycle phase, turning points, other….)
  - **Lead/lag analysis** between credit and business cycles (DTW / cross-correlation)
  - **Forecasting inputs** for XGBoost / PyMC/Other:
    - macro variables
    - credit spreads
    - PD shifts / early warning indicators

- **Other Latent-Variable Methods:**
  - Dynamic Factor Models(Analysis)
  - Kalman Filters & State-space models (filtered vs smoothed; linear / nonlinear)
  - Denoising Autoencoder
  - Etc.